

Question Paper

Exam Date & Time: 24-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: CIV 1003/CIV_1003 ENGINEERING MECHANICS AND SMART BUILDINGS

ENGINEERING MECHANICS AND SMART BUILDINGS [CIV 1003]

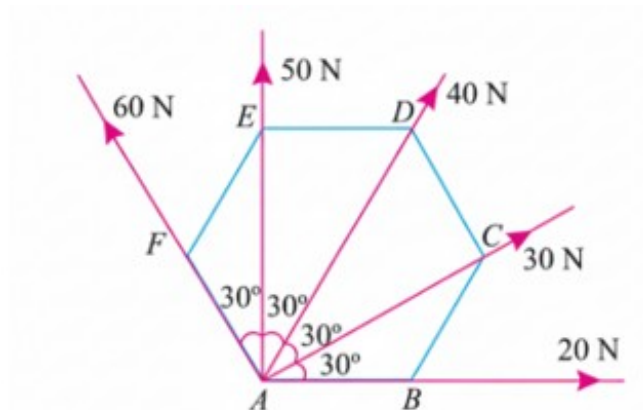
Marks: 50

Duration: 180 mins.

A

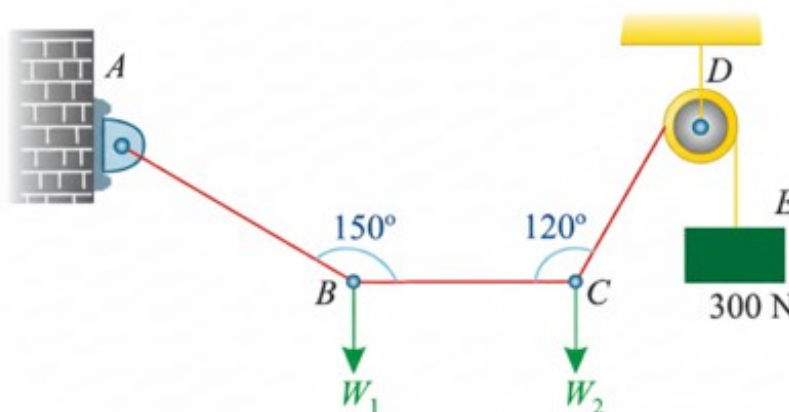
Answer all the questions.

- 1A) The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon, towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force. (5)



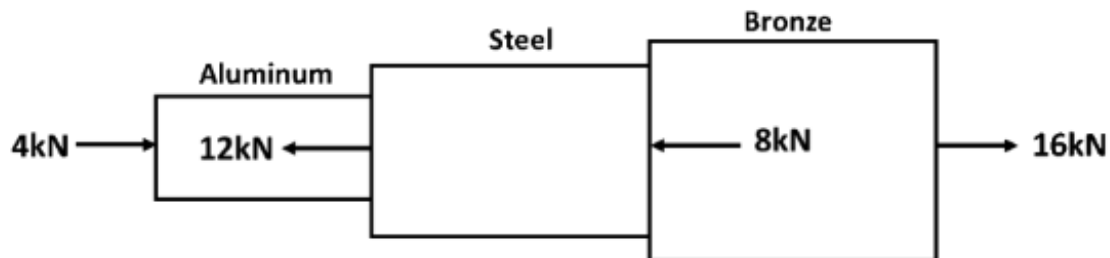
Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) A light string ABCDE whose extremity A is fixed, has weights W_1 and W_2 attached to it at B and C. It passes round a small frictionless pulley (neglect the weight of pulley) at D carrying a weight of 300 N at the free end E as shown in the figure. If in the equilibrium position, BC is horizontal, find (5)
- (i) Tension in the portions AB, BC and CD of the string and
- (ii) Magnitudes of W_1 and W_2 .



Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) Explain the Energy Efficiency Ratio (EER) and the Seasonal Energy Efficiency Ratio (SEER) (5)
highlighting the differences between their operating conditions.
- 2B) Calculate the stress developed in each section of the composite bar consisting of steel, aluminum, and (5)
bronze sections rigidly connected as shown in the figure. Also, determine the deformation of each section
and the overall change in length of the bar.
Given:
 $E_{\text{bronze}} = 100 \text{ GPa}$, $E_{\text{aluminium}} = 70 \text{ GPa}$, $E_{\text{steel}} = 200 \text{ GPa}$.
Cross sectional area and length of Aluminum section: 600 mm^2 and 800 mm respectively.
Cross sectional area and length of Steel section: 1000 mm^2 and 1200 mm respectively.
Cross sectional area and length of Bronze section: 1500 mm^2 and 1400 mm respectively.



- 3A) A steel bar $40 \text{ mm} \times 40 \text{ mm}$ in cross-section and 140 mm length is subjected to an axial (5)
compressive load of 250 kN longitudinally.
i) Determine the deformation in length of the bar if the lateral strain is fully prevented.
ii) Determine the deformation in length of the bar if only one-third of the lateral strain is prevented.
Take Young's modulus and Poisson's ratio of steel bar as 205 GPa and 0.3 respectively.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) Derive the relationship between young's modulus (E) and modulus of rigidity (G) for a deformable (5)
body.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4A) A mild steel bar of 30 mm diameter, 250 mm length, has an extension of 0.20 mm under a load of (5)
 85 kN . The load at the elastic limit is 170 kN , and the ultimate load is 280 kN . The total extension is
 45 mm and diameter at the fracture is 19.5 mm . Calculate the elastic modulus of the material, stress
at the elastic limit, ultimate strength of the material, percentage elongation and percentage
reduction in cross-sectional area.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) A bar 25 mm in diameter and 500 mm in length is acted upon by an axial load of 40 kN . The (5)
elongation of the bar and change in diameter are measured as 0.175 mm and 0.0041 mm ,
respectively. Calculate Poisson's ratio and three moduli. Also, determine the allowable stress if a
factor of safety is 1.5 .

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Explain briefly five types of displacement sensors. (5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) Explain the working mechanisms of (5)
i) Strain gauge
ii) pH sensor

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CIV_1003:ENGINEERING MECHANICS AND SMART BUILDINGS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	4	CO 4	20
4	5	CO 5	10
Total			40

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CIV_1003:ENGINEERING MECHANICS AND SMART BUILDINGS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	15(37.5%)
3	Application	5	25(62.5%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		8	40 (100%)